

## COM410 Programming in Practice

### Case Study

#### Specification

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##### Motorway Cricket

Motorway Cricket is a vehicle-spotting game played by two teams. One person at a time comes up to 'bat', and they score 'runs' when vehicles are sighted. To count as runs, vehicles must be moving and traveling in the opposite direction. Points are awarded according to the type vehicles as follows.

| Vehicle Type | Points |
|--------------|--------|
| Car          | 1      |
| Van          | 4      |
| HGV          | 4      |
| Bus          | 6      |
| Minibus      | 6      |

The player continues to accumulate points until any red vehicle is counted. A red vehicle scores no points but ends that player's turn. In a match, the players in a team play in series, one after another, with their accumulated score recorded. When all players in the first team have completed, the players of the second team build their scores in the same way. When all players of both teams have completed, the team with the highest total score is declared the winner.

##### Task

Implement a simulated game of Motorway Cricket for two teams of three players each. Generate a stream of random vehicles according to the following distributions.

| Vehicle Type | Percentage |
|--------------|------------|
| Car          | 75%        |
| Van          | 18%        |
| HGV          | 4%         |
| Bus          | 2%         |
| Minibus      | 1%         |

| Vehicle Colour | Percentage |
|----------------|------------|
| Gray           | 26%        |
| Black          | 20%        |
| White          | 17%        |
| Blue           | 16%        |
| Red            | 9%         |
| Silver         | 7%         |
| Green          | 2%         |
| Orange         | 1%         |
| Yellow         | 1%         |
| Bronze         | 1%         |

The only user involvement in the simulation is to provide the names of the players in each team. As the simulation proceeds, it should report each vehicle spotted and the points awarded, the total scored by each player, and the total number of runs for the team. At the end of the game, it should present the final score and declare the winning team.